PHINMA Araullo University - South
College of Information Technology
Bachelor of Science in Information Technology
S.Y. 2025-2026 | 2nd Semester

ITE 401 Platform Technologies—Sprint 1 | Period 1 Requirements

Simple Cryptocurrency Wallet
Preliminary Documents

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ITE 401 Platform Technologies—SPRINT I | Period 1 Requirements
Simple Cryptocurrency Wallet | Preliminary Documents

I. Project Name and Description

CoinNest is a simple web-based cryptocurrency wallet that lets users easily set up a wallet, check their balances, and send and receive crypto with just a few clicks. It puts a lot of emphasis on making sure users understand how to use the service by providing guided onboarding, copy-safe receive addresses, and clear network fees before confirmation. CoinNest's blockchain layer and modular web architecture are not tied to any one provider, so it can grow to support more coins and chains without having to change the user interface. It meets business needs by making it easier for new customers to get started and paving the way for future integrations like swaps and fiat on ramps.

II. Problem Statement

Many new and casual crypto users have trouble keeping their digital assets safe since current wallets can be hard to use, make mistakes, and require too much work for simple things like sending or receiving money. This results in expensive errors (incorrect addresses, unforeseen charges), inadequate security practices, and diminished trust—particularly for customers seeking a simple method to manage and transfer cryptocurrency. Minor retailers and online vendors have obstacles in payment acceptance due to intricate configurations and ambiguous transaction statuses. CoinNest mitigates this issue by offering a streamlined web wallet featuring guided onboarding, transparent confirmations, and explicit transaction tracking, thereby enhancing the safety and convenience of daily cryptocurrency usage.

III. Feature Pyramid

Base Layer: Reliable Wallet Basics

Feature 1. Guided Onboarding with Pro-Grade Key Control
New users can create or import a wallet with step-by-step guidance, simple definitions, and a short safety checklist. Crypto professionals get clearer key visibility (what's stored where), optional advanced backup methods, and a clean "security status" view that confirms the wallet is set up correctly.

Feature 2. Clean Send/Receive with Strong Error Prevention
Beginners can send and receive with QR scan, copy/paste, and a friendly review screen that highlights the most important details (asset, network, address, amount, fee). Pros get faster flows—saved contacts, address book labels, and one-tap "repeat send" from transaction history—without sacrificing confirmation checks.

Middle Layer: Clarity, Tracking, and Control

Feature 3. Portfolio + Transaction Insights
Beginners see an easy overview: total value, major holdings, and recent activity, with minimal noise. Professionals can drill down into per-asset performance, exportable transaction records, and detailed receipts (Tx hash, fee breakdown, confirmation count) for audits or reporting.

Feature 4. Network & Fee Management
Beginners choose from "Low / Standard / High" fees with plain-language tradeoffs and ETA ranges. Professionals can toggle advanced controls, custom fee, nonce/replacement options (where applicable), and clear network selection while the wallet actively prevents mismatched network mistakes.

Top Layer: Security, Trust, and Recovery

Feature 5. Security Center + Recovery Confidence
The wallet provides biometric/PIN lock, auto-lock, scam/mistake warnings, and a "Security Center" that shows what's protected and what still needs attention. Beginners

get a guided recovery plan and periodic backup reminders, while professionals can enable stricter controls (e.g., transaction confirmation friction, privacy mode, and stronger authentication options).

IV. User Story

Base Layer: Reliable Wallet Basics

Feature 1. Guided Onboarding with Pro-Grade Key Control

As a **beginner learning crypto**, I want the wallet to explain setup in simple terms and help me back up safely, so I can start without feeling lost or making a risky mistake. As a **crypto professional**, I want transparent, controllable key management and a clear security status, so I can trust the wallet for serious funds and workflows.

Feature 2. Clean Send/Receive with Strong Error Prevention

As a **beginner**, I want to send and receive using QR codes and a clear "review before send" screen, so I don't accidentally send to the wrong place. As a **professional who makes frequent transactions**, I want faster repeatable flows and saved contacts, so I can move funds efficiently while still catching critical errors.

Middle Layer: Clarity, Tracking, and Control

Feature 3. Portfolio + Transaction Insights

As a **beginner**, I want a simple dashboard that shows my balance and recent activity, so can understand what's happening without digging through details. As a **professional**, want detailed transaction data and export options, so I can reconcile activity, document transfers, and support compliance or accounting needs.

Feature 4. Network & Fee Management

As a **beginner**, I want the wallet to recommend sensible fees and warn me if I'm on the wrong network, so I don't overpay or lose funds. As a **professional**, I want the ability to fine-tune fees and network settings when necessary, so I can optimize cost, speed, and reliability for different situations.

Top Layer: Security, Trust, and Recovery

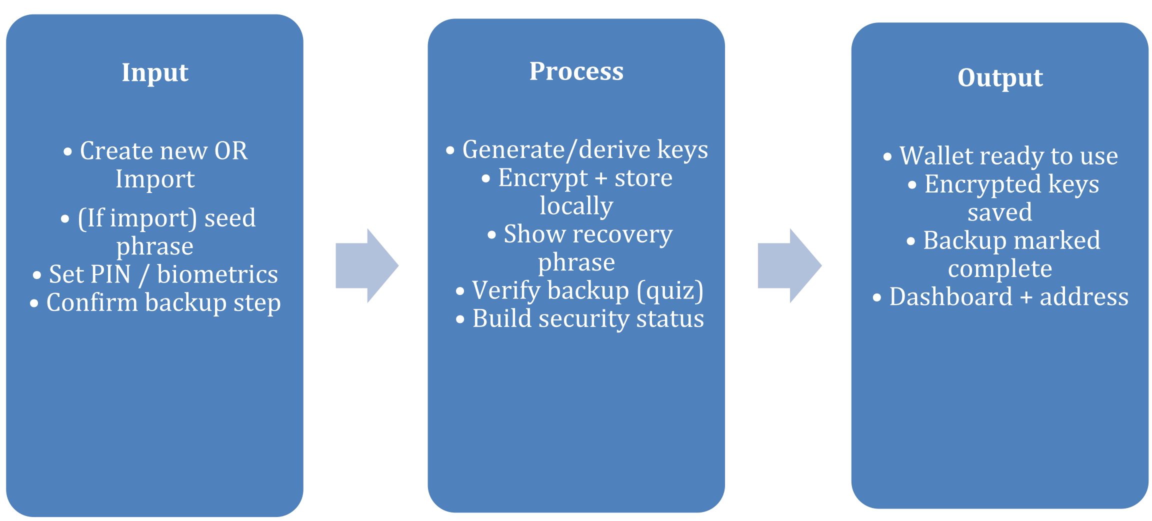
Feature 5. Security Center + Recovery Confidence

As a **beginner**, I want a straightforward recovery plan and reminders to store my backup safely, so I don't lose access if my phone is lost or replaced. As a **crypto professional**, I want stronger security controls and a clear overview of protections, so I can confidently use the wallet as part of a high-stakes routine.

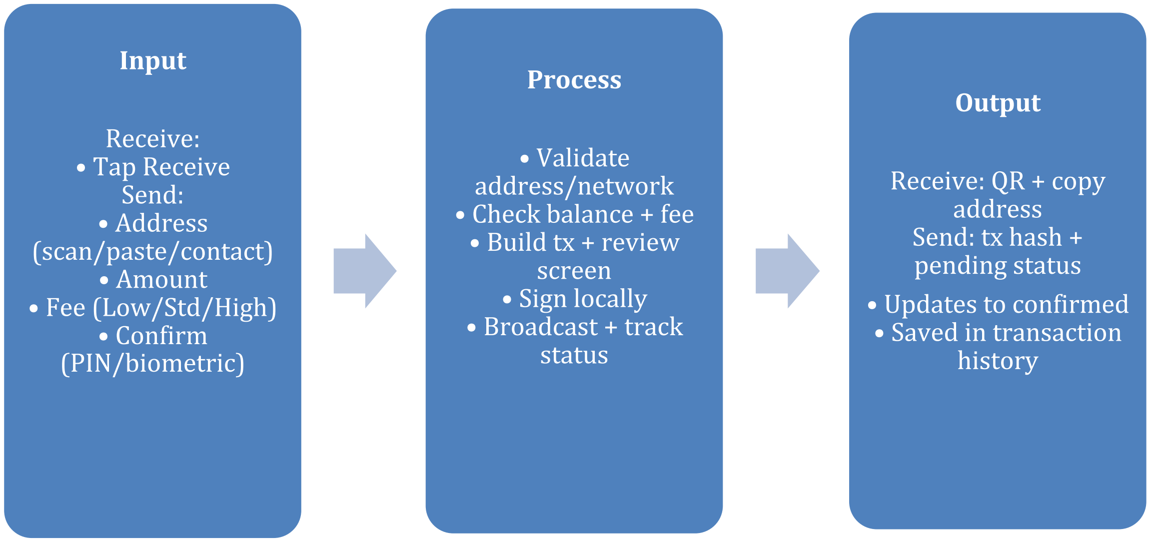
V. Flow Diagram

Base Layer: Reliable Wallet Basics

Feature 1: Easy setup that still feels secure

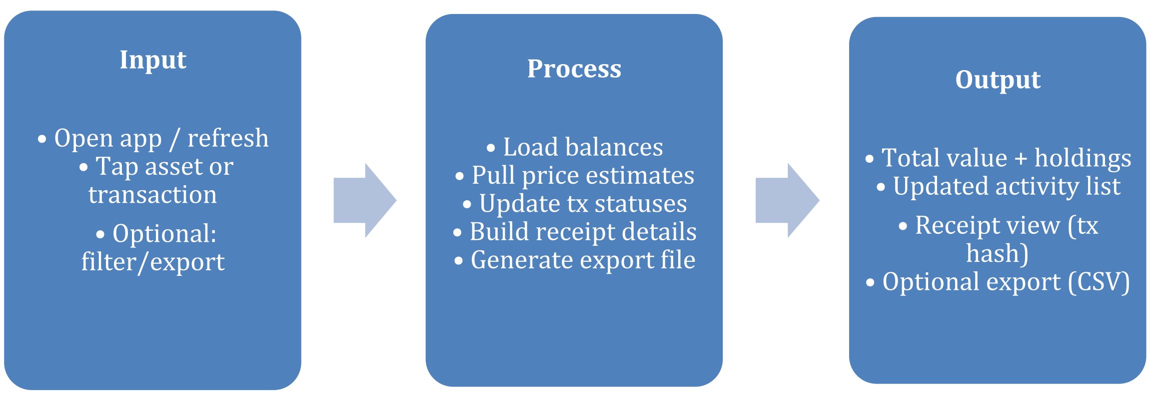


Feature 2: Sending and receiving that doesn't make you nervous

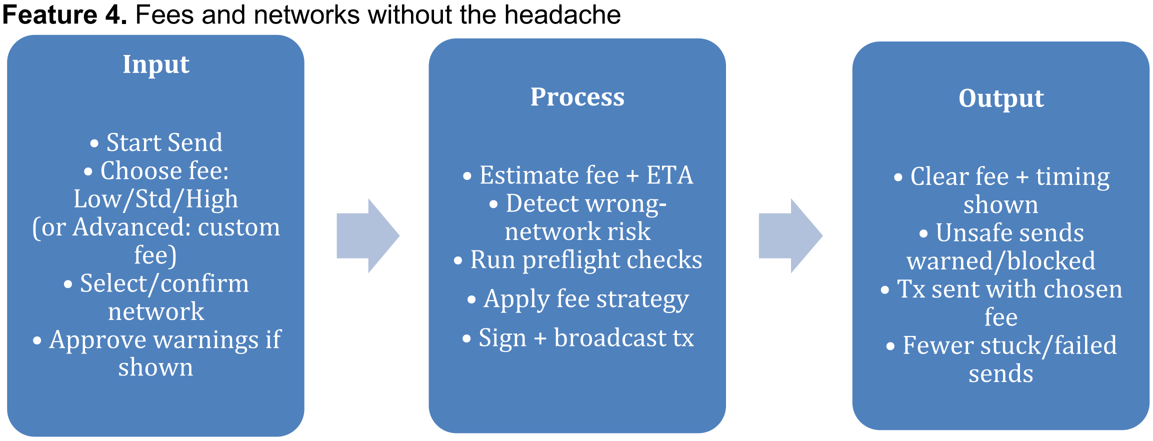


Middle Layer: Clarity, Tracking, and Control

Feature 3: Dashboard that's simple until you need details

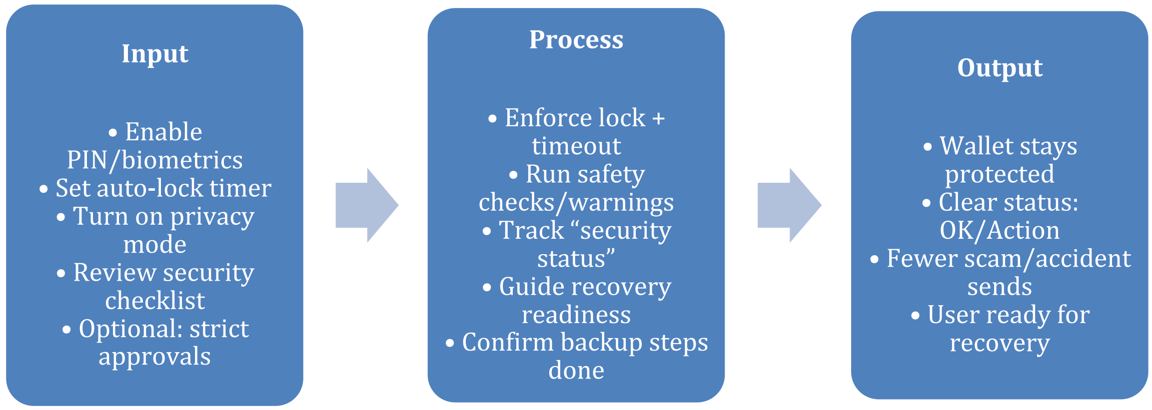


Feature 4. Fees and networks without the headache



Top Layer: Confidence & Safety

Feature 5. Security Center that keeps you out of trouble



VI. Roles and Responsibilities

Role	Responsibilities	Assigned Team Member
Project Manager	Organizes tasks, timelines, and meetings; tracks progress and removes blockers to keep the team on schedule.	**Harold Harvey C. Cambe**
Frontend Developer	Builds the wallet UI (send/receive, balances, history), connects to Web3 providers, and ensures a smooth user experience.	**Fidel A. Gonzales III.**
Backend /API Developer	Integrates price feeds and network data (fees/ETAs), manages optional services (notifications), and supports export/report features.	**Jetheo Christian S. Ocampo**
QA	Integrates price feeds and network data (fees/ETAs), manages optional services (notifications), and supports export/report features.	**Ace DC. Arcilla**
UI/UX	Integrates price feeds and network data (fees/ETAs), manages optional services (notifications), and supports export/report features.	**Emmanuel Isaiah J. Javate**

VII. Development Timeline

Week Number	Activities, Tasks, and Deliverables	Date
Week 1	**Activity:** Environment Configuration **Tasks:** Open Remix IDE • Install OpenZeppelin library imports • Configure workspace **Deliverable:** Ready-to-use dev environment	**Feb 9–13**
Week 2	**Activity:** Smart Contract Foundation (Token + Basic Wallet Logic) **asks:** Set up ERC-20 template (if needed) • Implement core functions • Add events • Write initial unit tests **Deliverable:** Deployed/tested contract on local test network + test suite v1	**Feb 16–20**
Week 3	**Activity:** Wallet Core Features (Send/Receive + UI Skeleton) **Tasks:** Build send/receive screens • QR scan/generate • Address validation • Connect to testnet	**Feb 23–27**

	• Basic transaction status (pending/confirmed) **Deliverable:** Working prototype: send/receive on testnet with confirmations	
Week 4	**Activity:** Visibility & Controls (Dashboard + Fees/Networks) **Tasks:** Portfolio view + transaction history • Fee presets (Low/Std/High) • Network selection & wrong-network warnings • Receipt details (tx hash, fee) **Deliverable:** Feature-complete MVP flows (beginner + pro toggle options)	**Mar 2–6**

VIII. Risk and Mitigation Table

Risk	Mitigation
The admin account is lost or compromised.	Apply **Principle of Least Privilege**: admin/dev roles can only pause/upgrade, never access user funds. Use **multisig** for admin actions and store keys in **hardware wallets** with secure backups.
User loses their seed phrase / recovery phrase.	Make backup unavoidable during onboarding (clear steps + confirmation quiz). Add a short reminder to verify their backup and explain recovery plainly: **no seed phrase = no recovery**.
User sends funds to the wrong address.	Use QR scanning, address validation/format checks, and a strong “review before send” screen. Warn on first-time addresses and add an extra confirmation for unusually large amounts.
User sends on the wrong network (chain mismatch).	Show the selected network clearly, auto-detect when possible, and warn/block mismatched networks. Provide a simple explanation of what can go wrong before letting them proceed.
Smart contract bug or vulnerability.	Rely on audited libraries (OpenZeppelin), write unit tests for every critical function, run basic static analysis, and do at least one peer code review before deploying beyond testnet.

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PHINMA EDUCATION

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PHINMA EDUCATION

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Date : 25/2/2026

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ITE 293	AU-BSIT3-S2	TBA,408 Phase 2	Gabriel Nikiloz Guinto Sioson	
ITE 370	AU-BSIT3-S2	TBA,407 Phase 2	John Frederick Leabres	
SSP 008	AU-BSIT3-S2	TBA	Salvador Collado	
ITE 235	AU-BSIT3-S2	402 Phase 2,407 Phas	Matthew Padilla Corpuz	
ITE 386	AU-BSIT3-S2	402 Phase 2,407 Phas	Aira Lucero	
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